SURBHI SHIRVATKAR

Mumbai, India | surbhi1us@gmail.com | LinkedIn | GitHub

EDUCATION

Mukesh Patel School of Technology Management & Engineering, NMIMS
B. Tech in Data Science | 2024–2027
Diploma, Computer Science | 2021–2024

Cambridge School
10th | ICSE | 2021

TECHNICAL SKILLS

Programming: Python, SQL, C++
Data Science & Machine Learning: Pandas, NumPy, Scikit-learn, TensorFlow, Keras
Data Visualization: Power BI, Tableau, Plotly
Development & Tools: FastAPI, REST APIs, Streamlit, Git, GitHub
Databases: MySQL, SQLite, Firebase
Cloud: AWS (EC2, S3, IAM)

PROJECTS

Smart Stock Analyzer | Python, Pandas, Streamlit, Plotly, yfinance
• Developed an interactive stock-analysis application using live market data to evaluate equities across return, annualized volatility and risk-adjusted performance using the Sharpe ratio.
• Implemented objective-based stock ranking for maximum return, minimum risk and balanced strategies, with correlation analysis and interactive Plotly visualizations.
• Built Monte Carlo GBM simulations to generate P10/P50/P90 price scenarios and integrated sentiment analysis and an AI assistant for interactive financial analysis.

Image Denoising using U-Net | Python, TensorFlow/Keras, FastAPI, AWS
• Developed a custom U-Net with skip connections, batch normalization and dropout for Gaussian and Poisson image denoising, achieving 0.75 SSIM on BSDS500.
• Built a FastAPI REST endpoint for image inference and deployed the end-to-end web application on AWS EC2.

Nationwide Agriculture Production Forecasting | Python, Pandas, StatsModels, Auto-ARIMA
• Processed 40+ years of Indian crop-production data across states and crops and developed automated forecasting pipelines using Holt's Linear Trend and Auto-ARIMA.
• Applied KPSS stationarity testing to guide model configuration and generated 10-year State × Crop forecasts with aggregated All-India production outlooks.

Smart Traffic Signal System | Python, YOLOv8, OpenCV
• Developed an AI-based traffic system using YOLO models for vehicle, emergency-vehicle and orientation detection.
• Implemented IoU-based matching and dynamic signal timing using lane-wise vehicle counts, with priority handling for emergency vehicles.

CERTIFICATIONS

Deloitte Australia — Data Analytics Job Simulation, Forage
Cisco Networking Academy — Introduction to Modern AI
SAP ERP Introductory Course — SAP MM, SD & ABAP | 90 Hours

ACHIEVEMENTS & LEADERSHIP

- **2nd Place — ADAPPT Research Event, IETE-SF / Amazon / NMIMS (2022)** — Developed *Cyber Rakshak*, a digital-safety application.
- **Smart India Hackathon 2025** — Proposed an AI-driven public-health chatbot.
- **Volunteer** — We Can We Will Foundation